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	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
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(1) Research Program/Project/Study Title: Development of Tray Bag as Culture Method for Oyster	
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Program or Project has component (if no proceed to **2**)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation: Pilar Bay (Panay, Pontevedra, President Roxas and Pilar) Capiz	
Municipality/City:	
Province:	Capiz
Region:	IV
(5) Classification Research: ☐ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Oyster	(8) Discipline/Field of Specialization

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(9) Rationale

Oyster farming is among the highly profitable industries in Southeast Asia. Oysters provide a good source of livelihood and cheap source of food. Their affordability to low-income members of the society makes them among the most in-demand bivalves in local markets. In the Philippines, oysters are considered as one of the most valuable bivalve mollusks.

The earliest available record on oyster culture, employing the very crude broadcast method where oysters were manually removed from cultches, broadcasted on subtidal areas along estuaries and left to grow until harvest size, dates back to as early as 1931 in Hinigaran, Negros Occidental (Rosell, 1991). Few years later, the use of bamboo stakes was adopted (Delmendo, 1989), and though not environmentally friendly as it promotes bio-deposition of wastes on the seabed (Bacalso, 2014) it is still popular. Together with clams, scallops, cockles, abalones and mussel which are the most cultured mollusc, oyster has been one the predominantly culture bivalve in the country. In fact, Philippines produced over 53 thousand metric tons (MT) of oysters in 2020, with its value reaching P1 billion. Bulacan is the top producer with a total of over 53 thousand MT. This is followed by Capiz with oyster production of 8 thousand MT and 1.4 thousand MT.

Although, despite the decade old of practice of culturing these mollusks and in producing tons of harvest, the methods of growing oyster remained traditional. As reported by Angell (1986) and Sitoy (1988), the substrate used for collecting spats remained the same, oyster farmers still use bamboo, tires, oyster shells, and ropes. Although from broadcast and bamboo stakes or tulos methods, other methods have evolved through the years, such as the hanging, long line and lattice methods (Delmendo, 1989). Regardless of the type of substrate and culture method used, these are just left in the culture area for the spat to settle, grow into marketable size and later harvested. Nevertheless, these small improvements in culture methods produced the same quality of oysters as before—clumping and with a wide range of sizes. Although better and newer methods are available in other countries, like the pouch method in Thailand (Tanyaros et al., 2008), the Stanway cylinders in France (Robert et al., 1993) and the stick, tray, basket, cylinder and other sophisticated methods in Australia (Leavitt, 2017) oyster growers in the Philippines still prefer to use the traditional methods learned from their ancestors (Lebata-Ramos et al., 2021).

Using the traditional methods, harvested oysters clustered or clumped together, vary in a wide range of sizes, and in the usual farming harvested and sold when they attain a mean shell length of 60–90 mm (Lovatelli, 1988). Oysters cultured and gathered through these traditional methods often end up being sold locally, primarily in coastal markets or small roadside stalls. On the other hand, high-end hotels and restaurants prefer imported oysters, which are individually separated and consistently sized. Local producers aiming to access this profitable market must undergo a painstaking process of separating the oysters from their clusters, cleaning them, and individually sorting them to meet the size standards demanded by these establishments. The need for regular culling during the initial stages of cultivation, followed by the laborious task of individually separating clumped oysters during harvest, requires significant manpower, additional resources and is time consuming. Therefore, to boost production and improve oyster quality, other culture systems and new technologies must be explored to support the growing market and the increasing demand for higher-quality oysters.

Moreover, the present and traditional practices employed in oyster farming pose environmental concerns due to the use of bamboo stakes. These stakes have been observed to trap sediments, leading to sedimentation, which ultimately contributes to the shallowing of bays. Sedimentation which emerged as a pressing concern which has implications in the biodiversity and habitat quality.

Thus, this study will aim to modify and develop tray bag culture method for oyster that will produce individual or single oyster and a culture method/technique that will minimize siltation and its impact on the marine environment.

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(9-A) Project Description

This study will apply the development type of research, this proposed study intends to help in the production and improve quality of oyster to meet the increasing demand for high-quality oyster. The development of tray bag for oyster culture will also aim to help address and minimize the problem in siltation.

(10) Objectives/Statement of the Problem

Generally, this study aims to develop and construct the tray bag as a culture method for oyster.

Specifically, it aims to:

1. Construct the tray bag to be used as the culture method for oyster
2. Evaluate the tray bag as a culture method in terms of survival rate
3. Evaluate the tray bag as a culture method for oyster growth rate in terms of:
 - a. size
 - b. weight
 - c. Meat yield

(11) Scientific Basis

KR101281028B1 oyster culture structure

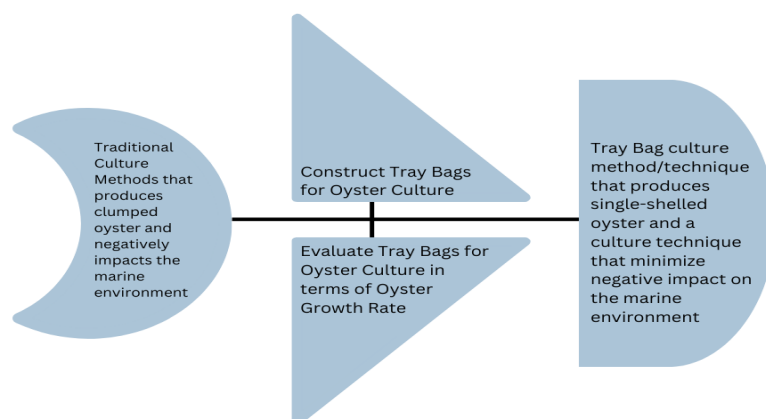
The present invention is an installation on the seabed that trains individual oyster spat. It has a simple structure, has low resistance to the flow of seawater, is durable, and has a training net that moves along the flow of seawater to collect the spat contained therein. It is about a structure for cultivating oysters that is good for vegetation and training. For this purpose, a pair of supports fixed to the seabed at a predetermined interval; A mounting portion connecting the pair of support portions; And it includes a bag-shaped benign net that is hung on the holder. Therefore, the structure for cultivating oysters according to the present invention is a simple structure that can hang a positive net, has low resistance to the flow of seawater, is strong in durability, and can be easily installed on the seabed, so manufacturing and installation costs are low. In addition, by installing a large number of training nets between a pair of supports, a large amount of spat can be placed and nurtured per unit area. These training nets are placed to move along the flow of seawater, which has a good effect on the vegetation and training of the spat contained therein.

KR101889652B1 APPARATUS FOR OYSTER CULTURE AND CULTURING METHOD OF OYSTER

The present invention relates to an oyster culture device and a culturing method for oysters. Specifically, the oyster culture device includes: a first aquaculture net connection unit in which two or more vertical bars partially embedded in the seabed are positioned in a straight line and spaced apart; a second aquaculture net connection unit partially embedded in the seabed, wherein the two or more vertical bars are positioned to face the two or more vertical bars of the first aquaculture net connection unit at intervals of 110% or greater of the width of an aquaculture net; a first aquaculture net support unit having two or more horizontal bars connecting the vertical bar of the first aquaculture net connection unit and the vertical bar of the second aquaculture net connection unit; a second aquaculture net support unit having two or more vertical bars connecting the horizontal bar of the first aquaculture net support unit to each other; and the aquaculture net positioned in the first aquaculture net support unit and the second aquaculture net support unit and having a connection hole at both ends, wherein the aquaculture net is fixed by being individually inserted into the vertical bar of the first aquaculture net connection unit and the vertical bar of the second aquaculture net connection unit facing the vertical bar of the first aquaculture net connection unit. The oyster culture device can remarkably reduce labor for performing a culturing process in comparison with existing oyster culture devices and can produce a large quantity of good quality oysters. Also, the oyster culture device enables a user to easily collect the aquaculture net after the culture is completed.

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Theoretical Framework



(13) Methodology

Construction of the Tray Bag

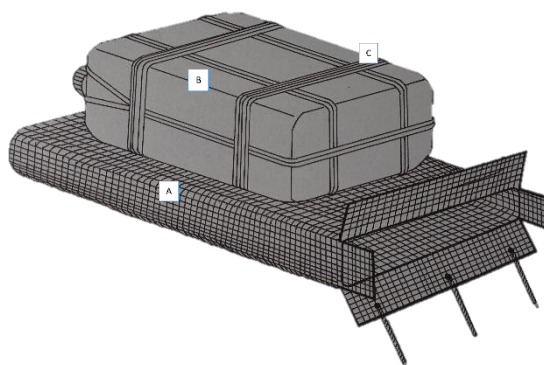


Figure 1. Model of the Tray Bag for Oyster Culture Method (a) tray bag made of green screen (b) black plastic container that will serve as the floater (c) Polypropylene and Polyethylene Rope for tying

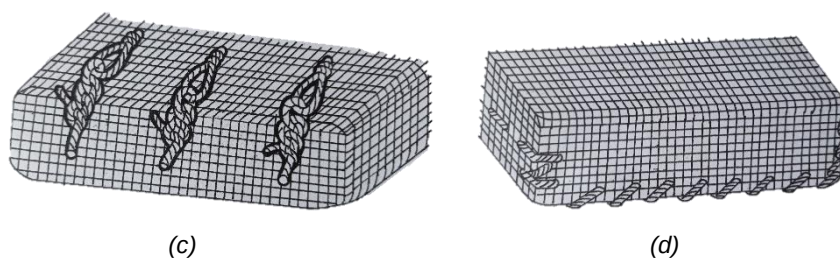


Figure 2. Tray Bag for Oyster Culture (c) opening of the tray bag and (d) end/back portion of tray bag

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Materials and Method

In constructing the tray bag, the materials that will be used are green screen wire, black plastic container and polypropylene and polyethylene rope for tying.

The tray bag will be designed as a culture method that produce single/individual shell oyster. It will be designed to float centimeters below the surface water to minimize the entrapment of sediments at the bottom resulting to minimize siltation. Construction process of the tray bag will be as follows:

1. The tray bag itself will be constructed using the 90 cm green screen wire, to form the bag, green screen will be measured, cut and folded into the exact measurements.
2. One end will be sewed using a PE rope to secure the bag, while the other end will serve as the opening of the tray bag.
3. At the opening of the tray bag, a rope will be knot tied to secure the tray bag.
4. At the top center portion of the bag, black plastic container will be attached to serve as the floater of the tray bag. It will be tied using a #10 PE rope.

Evaluation of the tray bag for oyster culture

Deployment of the Tray Bag

Constructed Tray Bag will be deployed in a culture site which suites the criteria for oyster culture. Three (3) replicates will be deployed in the culture site. Oyster spats will be sourced out and will be randomly selected for stocking, 100 spats will be stocked on each tray bag. Prior to stocking, each spat will be blotted dry, measured for shell length (mm SL) using a caliper and total body weight (g BW) using a digital weighing scale.

Monitoring for survival and growth rate

A total of 60 oysters will be randomly sampled and will be measured monthly for length and weight (20 from each replicate). Moreover, dead oysters will be checked to account for mortalities. Using the minimum size requirement of high-end markets as baseline for harvest, oyster in this experiment will be harvested when they attained an average body weight (ABW) of ≥ 85 g in any one of the culture replicates. After harvest, oyster in each tray bag will be removed and cleaned of fouling organisms, and each will be measured for its respective shell length and body weight. All dead oysters from each replicate will be noted to account for mortalities. Then, to determine the Meat yield (MT) from each replicate, 10% of the oysters will be randomly sampled. Each sample will be weighed, shucked, the whole oyster flesh will be removed, blotted dry with paper towel, and will be weighed (g). Meat yield (MT) will be determined using Bentazos-Vega et. al (2018) as:

$$MT = \frac{\text{fresh flesh weight}}{\text{body weight}} \times 100$$

Based on their shell lengths, oysters were also classified following the Pacific Oyster Grading System (ASCRC, 2009) as bistro (50–60mm), buffet (60–70 mm), standard (70–85 mm), large (85–100 mm) and jumbo (100–120 mm). This classification was used because there is no standard to classify oyster products in the Philippines.

Data Analysis

Data on monthly monitoring will be recorded and kept in an excel file. Monthly growth rates both for length (GR_L) and weight (GR_W), respectively, were obtained using the following equations:

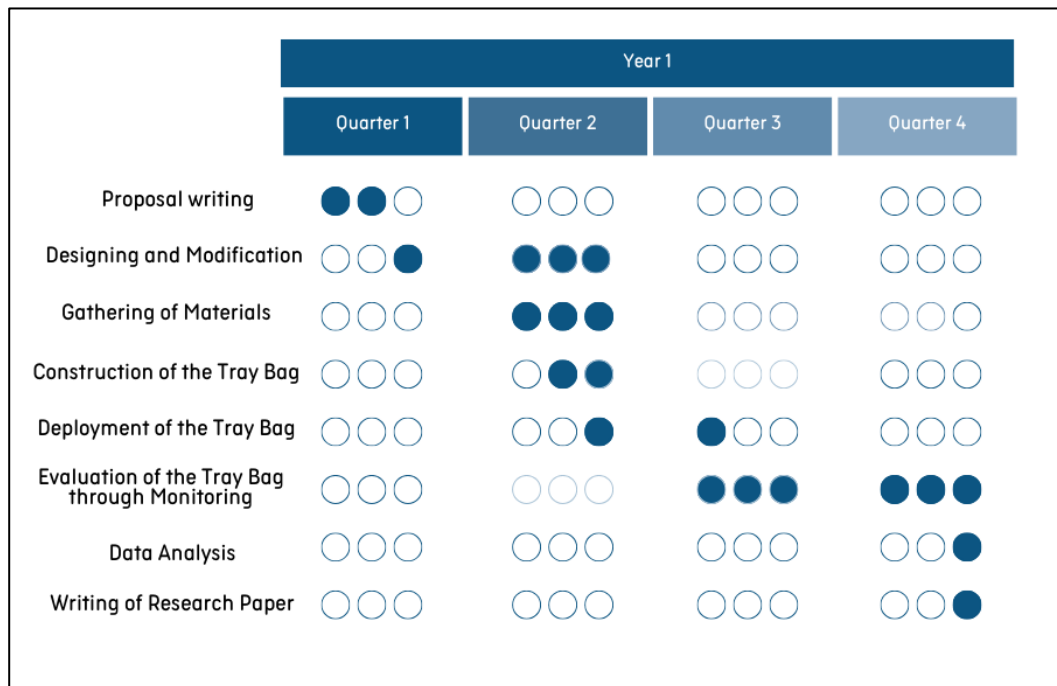
$$(a) \text{ GR}_L = \frac{L_f - L_i}{T_f - T_i} \times 30$$

$$(b) \text{ GR}_W = \frac{W_f - W_i}{T_f - T_i} \times 30$$

Where, GR, growth rate; L, length (mm); W, weight (g); T, sampling date; f, present sampling; i, preceding sampling; 30, constant, days in a month.

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(14) Major Activities/Work plan



(15) Expected Output

- Tray Bag as new culture metho/technology for production of single shelled oyster
- One (1) Patent/Copyright Application
- Extension services

(15) Technology Transfer Activity Design *(Indicate the specific products, activity design, processes or services to be transferred as extension activities)*

(16) Targets Beneficiaries

- **Oyster Farmers:** a new culture method/technology that will increase production with quality oyster, and increased Return of Investment because of minimal labor cost
- **Community:** will help address the environmental concerns on siltation

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study:

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement:

(19) Literature Cited

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)